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SSN_SEM: Design and application of a fusion ontology in the field of medical equipment

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Abstract

The importance of standardized operation of medical equipment is self-evident. An effective method is to construct the medical equipment domain ontology as the background knowledge base, and make judgments on the legality of the user's operation sequence based on the ontology, and guide the user to legally operate the equipment. After analysis, medical device data mainly comes from internal sensors. If we want to build an ontology background knowledge base, we can combine data with user operations, map the ontology to events, and build a directed graph model through events to form the final ontology. In this paper, we take the atomized particle preparation device in the field of medical equipment as an example, and selects and constructs the ontology by analyzing the internal structure and use process of the device. The application-level ontology selects the SSN ontology in the sensor field, and the upper ontology selects the general event ontology SEM. Combining the above two ontology and the characteristics of the medical device field, the ontology SSN_SEM is constructed, and the actual data of the device is semantically annotated through the SSN_SEM ontology, which verifies the validity of the ontology.

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Keywords: Medical equipment; domain ontology; event ontology

1. Introduction

Modern medical equipment is usually the crystallization of current human technology and wisdom. They are usually complex in structure and diverse in components. For example, a CT device usually includes a scanning

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module, a computer system module, and an image display and storage module. It can be said that a CT device is a combination of X-ray emission tube, small microcomputer, high-precision camera and high-definition display technology. The diversity of structures and functions of large-scale medical equipment means that its operations need to follow strict specifications and procedures. However, in the actual use process, it is inevitable that there will be situations such as illegal operations and those illegal operations may cause potential safety hazards. A series of management measures promulgated by the Ministry of Health clearly require that hospitals at all levels, especially hospitals above the second level, should strengthen the management of hospital medical equipment, improve the efficiency and management efficiency of medical equipment through informatization, and ensure the effectiveness and safety of medical equipment, And regular preventive maintenance of medical equipment^{1, 2}. Therefore, the early warning of illegal operation of medical equipment is a problem worthy of study. At present, effective results have not been achieved in the knowledge extraction of medical equipment. In order to achieve heterogeneous interoperability between multiple devices of the same type, and to improve the scientific and decision-making efficiency of the use of large-scale medical equipment, it is necessary to integrate and integrate medical equipment operating knowledge to achieve standardized operation of medical equipment. In this paper, by extracting equipment data, distilling the data into events, constructing the medical equipment ontology model, as a background knowledge base to guide users to legally operate the equipment.

2. Related Work

In the field of early warning system design for large-scale medical equipment, Chen Yan and others of Huai'an Second People's Hospital applied fuzzy neural networks to medical equipment to realize equipment alarms³; Xiong Gang and others of the Affiliated Hospital of Weifang Medical College The Internet of Things and single-chip microcomputer have designed a large-scale medical equipment early warning system, which is applied to the nuclear magnetic resonance equipment of our hospital³. Ontology in computer science is the formal expression of some concepts and their relationships to construct relevant knowledge involved in a field, which is the core technology of the Semantic Web. At present, combined with ontology as the background knowledge base, the related research on early warning of equipment violation operations is still lacking. If we want to use the ontology as the background knowledge base to realize the early warning based on the ontology, we first need to make a preliminary selection and construction of the ontology. Normally, almost every domain has a domain ontology suitable for its own usage scenarios. For example, in the face of massive sensor data, W3C proposed the concept of Semantic Sensor Network Ontology (SSN)⁵. SSN is the ontology used to describe a series of content such as sensors and their observation results. The properties, domains and description of SSN are shown in Table 1.

Table 1. Properties of SSN ontology.

Property	Domain	Description
sosa:hosts	sosa:Platform	What sensors does the platform have
sosa:isHostedBy	sosa:Sensor	Which platform does the sensor belong to
sosa:isObservedBy	sosa:ObservableProperty	Which sensor is the attribute observed
sosa:observes	sosa:Sensor	Properties of sensor observation
ssn:hasProperty	sosa:FeatureOfInterest	Properties of feature of interest
ssn:isPropertyOf	ssn:Property	Which feature does the property belong to
sosa:observedProperty	sosa:ObservableProperty	Property we observed
sosa:isFeatureOfInterestOf	sosa:FeatureOfInterest	Feature of interest
sosa:hasFeatureOfInterest	sosa:Observation	Feature of interest we observed
sosa:madeObservation	sosa:Observation	Make an observation action
sosa:madeBySensor	sosa:Sensor	Which sensor does the observe belong to
sosa:hasSimpleResult	xsd:dataTime	Result of observation
sosa:resultTime	RdfsLiteral	Timestamp of observation

We need to analyze data source, characteristics, and usage scenarios of medical equipment, mine semantic relationships from those datas, select and construct appropriate ontology. In addition, according to our needs, the constructed ontology needs to be used to describe each operation and the possible associations between them, and through these associations, the constraints on the illegal operation of the equipment are realized. Therefore, in addition to the field that is suitable for medical equipment usage scenarios, an ontology for describing the relationship between operations is also needed. If we want to build an appropriate ontology, we can find a suitable ontology based on the existing ontology to expand to meet our needs. This way can reduce the complexity of ontology construction.

3. Ontology Construction and Case Analysis

3.1. Ontology Construction

The first thing that should be determined is the domain and construction purpose of the ontology. The function of the application-level ontology is to receive the data generated by the device and map the data to events through semantic annotation. In order to design the application-level ontology, it is first necessary to analyze the equipment situation. we take the atomized particle preparation device in the field of medical equipment as an example for analysis. As shown in Fig. 1, the device is divided into five modules: high-frequency ultrasonic atomization module; multi-point dynamic monitoring module; efficient laminar electrostatic collection module; organic solvent recovery and inert gas circulation system; automatic control and data integration system. Automatic control and data integration system is our main source of data.

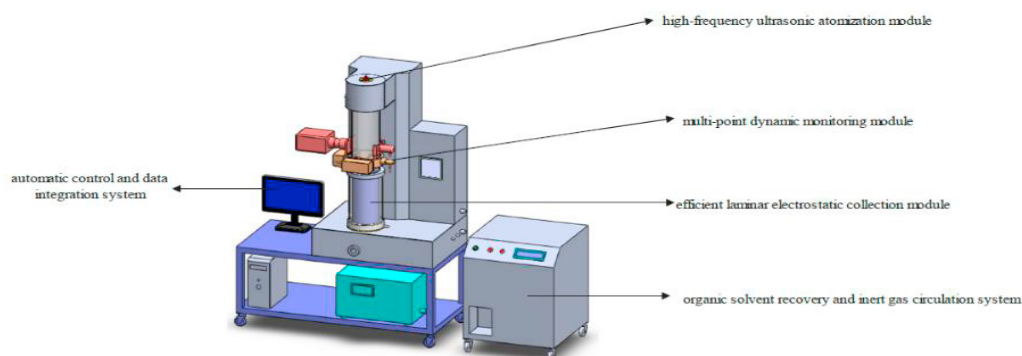


Fig. 1. Atomized Particle Preparation Device

Fig. 2.shows how to startup and use the device. According to Fig. 2., the main operations involved in the use of the equipment are: cooling, setting the oxygen content, closing the switch, turning on the fan, turning on the electric heating, turning on the injection pump and atomizing. Correspondingly, we install various of sensors. Since the data source is the sensors on the device, the ontology of the sensor field should be used as application-level ontology. Since the SSN ontology in the sensor field is already mature, we decided to use it.

Furthermore, since we need to judge the operating status of the equipment through a series of events of operating the machine, an ontology that describes events is also needed to serve as the upper-level ontology, because the events after the application-level ontology mapping lack semantic relations, and the upper-level ontology needs to attach semantic relations between these events. A general event ontology named Simple Event Model(SEM)⁶can be used by us. The core class of SEM is sem:Core, sem:Core contains four elements, namely sem:Event, sem:Actor, sem:Place and sem:Time, which represent what happened, the participants of the thing, the place where the thing happened and the time when it happened. The four elements correspond to different subclasses of sem:Type. In addition, we can also add constraints through sem:Constraint to them. It is a very versatile event ontology. Since our

requirement is to connect the events after the application-level ontology is mapped, SEM can fully meet our requirements. Part of the structure of SEM is shown in Fig. 3. We only used `sem:Event` among the four elements, because according to our needs, we only care about the user's operations on the device. These operations can be expressed as individuals of `sem:Event`, and we don't need to care about who the operator is, where is the operation and when is the operation time. Correspondingly, their subclasses of `sem:Type` are not required too.

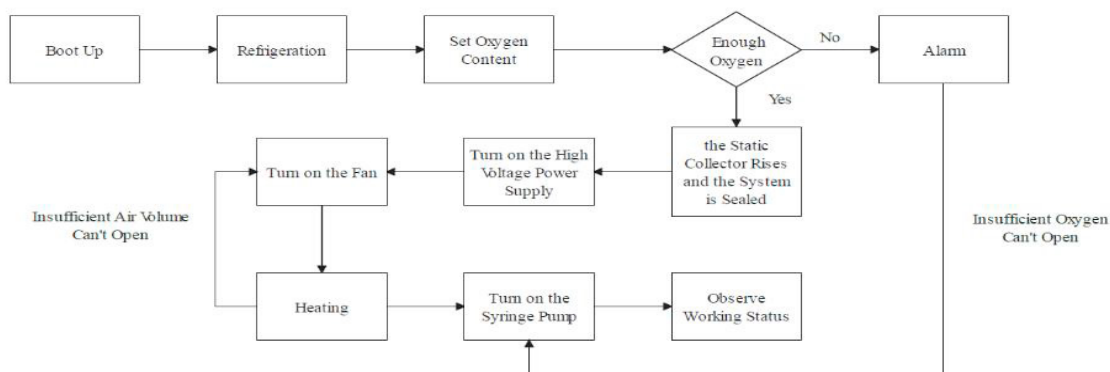


Fig. 2. Process of Startup

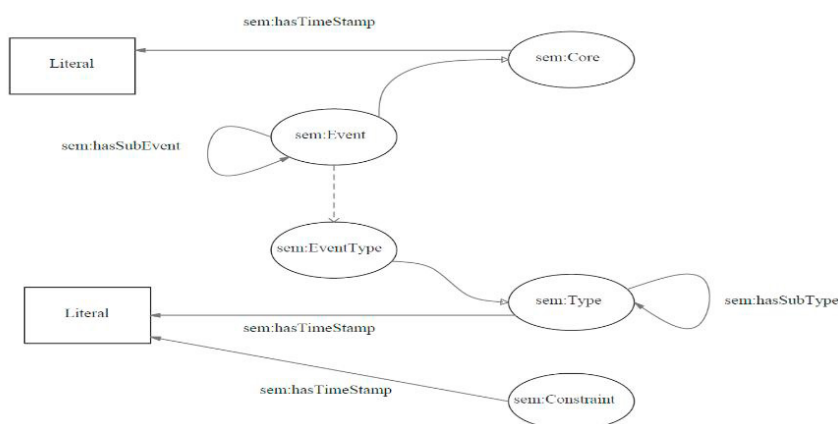


Fig. 3. Part of structure of Simple Event Model

After the above analysis, in the field of medical equipment, application-level ontology and upper-level ontology are all necessary for us. The data generated by the device is mapped by the SSN ontology at the application layer to form Observation or Actuations instances. Each instance actually corresponds to the `sem:Event` in the SEM. In other words, we can regard the `sosa:Observation` and `sosa:Actuation` classes in the SSN ontology as subclasses of the `sem:Event` class in SEM. Protégé is an ontology editing tool for Java. In protégé, we choose the Merge Ontologies option to merge the two layers of ontology to form a new ontology SSN_SEM. SSN_SEM can better meet our needs in the field of medical equipment. It can not only describe the data sampled by the sensor as an observation event or Actuation event, but also describe the relationship between different observation events or Actuation events. Only when the relationship between these events has a clear description, we can build a directed graph based on these relationships, so as to form a reasonable background knowledge base for early warning of illegal operation of medical equipment. The structure of SSN_SEM is shown in Fig. 4.

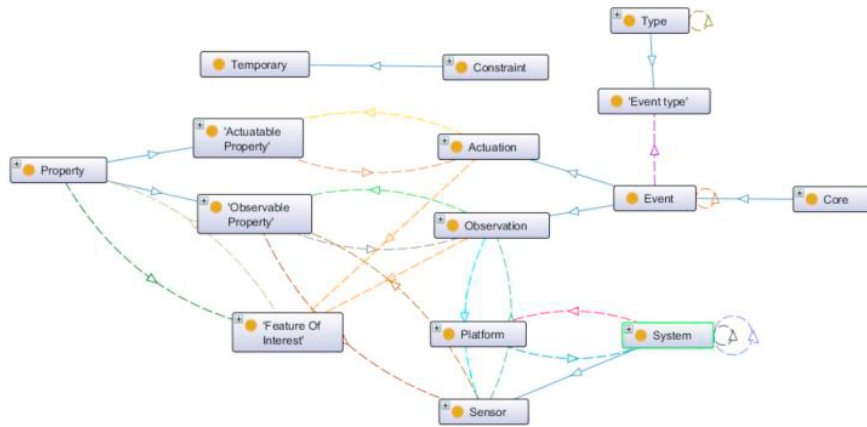


Fig. 4. Structure of SSN_SEM in protégé

3.2. Semantic Annotation

Semantic annotation needs to use the ontology parsing function of Apache Jena. Apache Jena is an open source Semantic Web framework for Java. It can be used to extract data from and write to RDF graphs with its API. Jena provides support for Web Ontology Language(OWL) so that we can use Jena to resolve ontologies. In addition, Jena also has various internal reasoners. The SSN_SEM obtained above is saved to disk in OWL format through the protégé tool, and the SSN_SEM ontology file is read using Jena to create the ontology model object OntModel. Obtain Sensor, Observation and Event class objects according to OntModel's getOntClass() method. For each piece of data received, we create instance objects for different class objects through our own createIndividual() method and XML mapping file information. After all the data is processed, use Jena to write the ontology file filled with the instance to disk.

3.3. Experimental Analysis

In order to verify the SSN_SEM ontology, we collected some data on the atomized particle preparation device for semantic annotation experiments. The experimental data comes from the instruments or sensors of different devices in the equipment automation control and data integration system. Sensors we used in our experiment are as follows: the SR80 temperature controller contains a series of temperature sensors; the oxygen concentration analyzer contains a concentration analysis device; a switch status indicator; the fan frequency converter contains a flow sensor; the MSP1-C2 industrial injection pump contains a speed sensor. Each row of the collected data is a record, and each record can be divided into six dimensions, as shown in Table 2 and The sample data structure collected by the sensor is shown in Table 3.

Table 2. Properties in the Collected Data.

Num	Property
1	Oxygen Concentration
2	Switch Status (close=1, open=0)
3	Voltage
4	Average Wind Speed
5	Heating Temperature
6	Injection Speed

Table 3. Sample Data Collected by Sensors

Oxygen Concentration	Switch Status	Voltage	Wind Speed	Heating Temperature	Injection Speed
0.218	1	220	75	143	2.5

We performed a simple semantic annotation experiment on these collected datas, and wrote the results to an OWL file. In order to check the logical correctness of the semantically annotated OWL file, we used protégé to open the OWL file and visualized class individuals and their relationships. The individuals and their relationship in protégé is shown in Fig.5. It can be seen that after processing by Apache Jena, the various attributes contained in the original sensor sampling data have been given clear semantic information. Thus verifying the logical correctness of the SSN_SEM ontology.

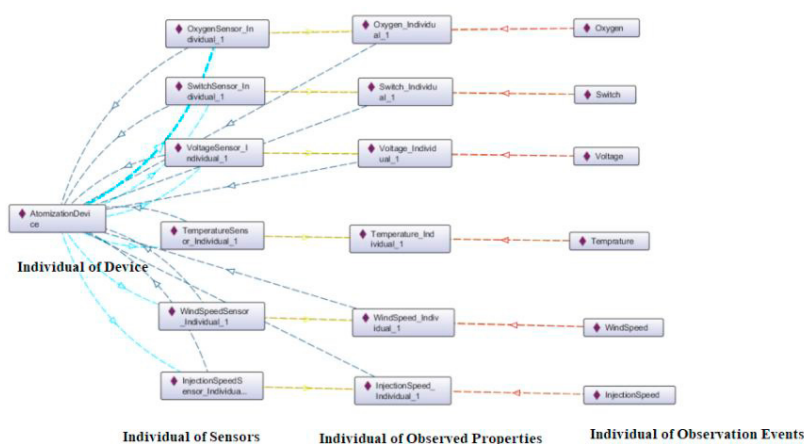


Fig. 5. Annotation Result in Protégé

4. Summary

This paper starts with the background of the early warning of the illegal operation of the atomized particle preparation device, we decide to use the SSN ontology as the application-level ontology through analyze data source,, and use SEM as the upper-level ontology through analyze our requirements. Through ontology fusion, we construct the SSN_SEM ontology for this project. After ontology constructed, the transformation from the original sensor data stream to the RDF event stream containing semantic relations is completed. Then, we save the RDF data as an OWL file and open it in protégé to verify the validity of SSN_SEM. However, there are many aspects that can be further studied. For example, the current semantic tagging system can only perform semantic tagging on a single ontology, and cannot support complex semantic tagging scenarios under multiple domain ontology. These issues need subsequent research and improvement.

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